

Generative AI Internship

Task 01: Text Generation with GPT-2

Submitted By:

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Internship:

Generative AI Virtual Internship

Organization:

PRODIGY InfoTech

1. Introduction:

Text Generation is one of the most important applications of Generative Artificial Intelligence. In this project, the GPT-2 pre-trained language model is used to generate meaningful and context-aware text from user-provided prompts. GPT-2 is capable of producing human-like text by predicting the next words based on the given input.

2. Objective:

The objective of this project is to implement text generation using the GPT-2 model and understand how transformer-based language models generate coherent and meaningful text from user prompts.

3. Tools and Technologies Used:

Python

Google Colab

Hugging Face Transformers

PyTorch

GPT-2

GitHub

4. Project Workflow:

Step 1

Open Google Colab and install the required libraries.

Step 2

Import the necessary Python modules.

Step 3

Load the pre-trained GPT-2 model and tokenizer.

Step 4

Provide a custom text prompt.

Step 5

Generate text using the GPT-2 model.

Step 6

Display and verify the generated output.

5. Implementation:

The GPT-2 model was loaded using the Hugging Face Transformers library. A custom prompt was provided as input, and the model generated meaningful text based on the context of the prompt. Different prompts were tested to observe the model's ability to produce coherent responses.

6. Results:

The GPT-2 model successfully generated context-aware text for different prompts. The generated responses demonstrated the effectiveness of transformer-based language models in text generation tasks.

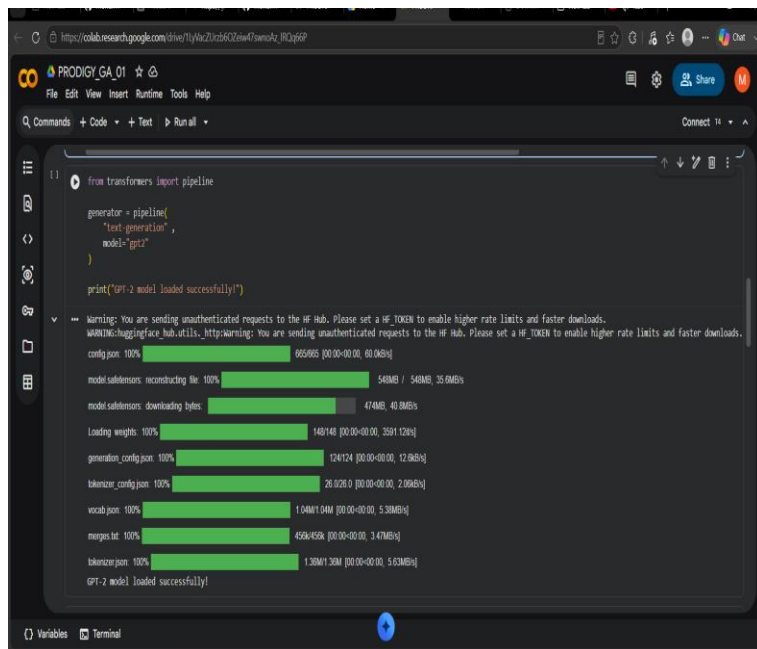
7. Output:

The notebook successfully generated text based on user prompts. Screenshots of the code execution and generated outputs are included in the GitHub repository.

```
[1]: !pip install transformers torch
```

-- Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.11.1)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.1.1+rocm@nvidia.com)
Requirement already satisfied: huggingface-hub<2.0.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.23.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex==2023.10.2 in /usr/local/lib/python3.12/dist-packages (from transformers) (2023.10.3)
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Requirement already satisfied: typing_extensions==4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.16.0)
Requirement already satisfied: sympy==1.02 in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: numpy==1.23.3 in /usr/local/lib/python3.12/dist-packages (from torch) (2.14.0)
Requirement already satisfied: networkx==3.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec==0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2023.3.0)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cuda,cudart,cufft,cutils,cupti,cusand,cusolver,cusparses,
Requirement already satisfied: cuda-bindings[CLBLAS]<12.8.1,>=12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cuda,cudart,cufft,cutils,cupti,cusand,cusolver,cusparses,
Requirement already satisfied: mdaia-core-cuda<9.18.0.6,>=9.18.0.6 in /usr/local/lib/python3.12/dist-packages (from torch) (4.9.0.5e)
Requirement already satisfied: mdaia-core-cuda<9.2.1 in /usr/local/lib/python3.12/dist-packages (from torch) (6.7.1)
Requirement already satisfied: mdaia-socv-cuda<2.8.3 in /usr/local/lib/python3.12/dist-packages (from torch) (2.8.9)
Requirement already satisfied: mdaia-nvcc-cuda<3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)
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Requirement already satisfied: mdaia-cblas-cuda<12.8.4.1,* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cudas,cudart,cufft,cutils,cupti,cusand,cusolver,
Requirement already satisfied: mdaia-cuda-runtime-cuda<12.8.*,* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cudas,cudart,cufft,cutils,cupti,cusand,cusolver,
Requirement already satisfied: mdaia-cuft-cuda<11.3.8.3,* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cudas,cudart,cufft,cutils,cupti,cusand,cusolver,
Note: you may need to restart your terminal to get the latest version of the package.

[illegible]



The first screenshot shows a Google Colab notebook titled 'PRODIGY_GA_01'. The code cell contains the following Python code:

```
from transformers import pipeline

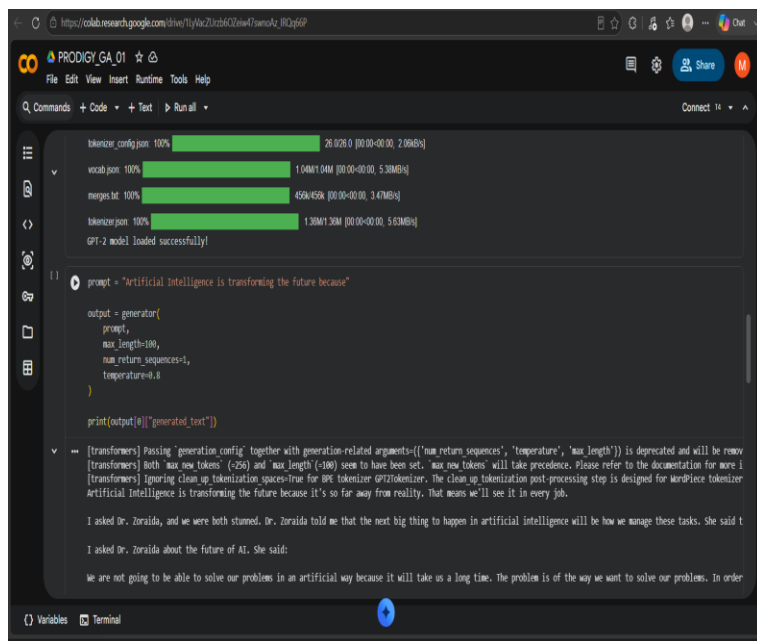
generator = pipeline(
    "text-generation",
    model="gpt2"
)

print("GPT-2 model loaded successfully!")
```

The output shows progress bars for various components of the GPT-2 model loading process:

- config.json: 100% (665/665 [00:00<00:00, 60.0kB/s])
- model.safetensors: reconstructing file: 100% (540MB / 540MB, 35.0MB/s)
- model.safetensors: downloading bytes: 474MB, 40.8MB/s
- Loading weights: 100% (140/140 [00:00<00:00, 3591.12kB/s])
- generation_config.json: 100% (124/124 [00:00<00:00, 12.0kB/s])
- tokenizer_config.json: 100% (26.0/26.0 [00:00<00:00, 2.09kB/s])
- vocab.json: 100% (1.04M/1.04M [00:00<00:00, 5.30MB/s])
- merges.txt: 100% (456k/456k [00:00<00:00, 3.47MB/s])
- tokenizer.json: 100% (1.30M/1.30M [00:00<00:00, 5.63MB/s])

The final output is: GPT-2 model loaded successfully!



The second screenshot shows the same Google Colab notebook. The code cell now contains the following Python code:

```
prompt = "Artificial Intelligence is transforming the future because"

output = generator(
    prompt,
    max_length=100,
    num_return_sequences=1,
    temperature=0.8
)

print(output[0]["generated_text"])
```

The output shows progress bars for the same components as the first screenshot, but with different values:

- tokenizer_config.json: 100% (26.0/26.0 [00:00<00:00, 2.09kB/s])
- vocab.json: 100% (1.04M/1.04M [00:00<00:00, 5.30MB/s])
- merges.txt: 100% (456k/456k [00:00<00:00, 3.47MB/s])
- tokenizer.json: 100% (1.30M/1.30M [00:00<00:00, 5.63MB/s])

The final output is: GPT-2 model loaded successfully!

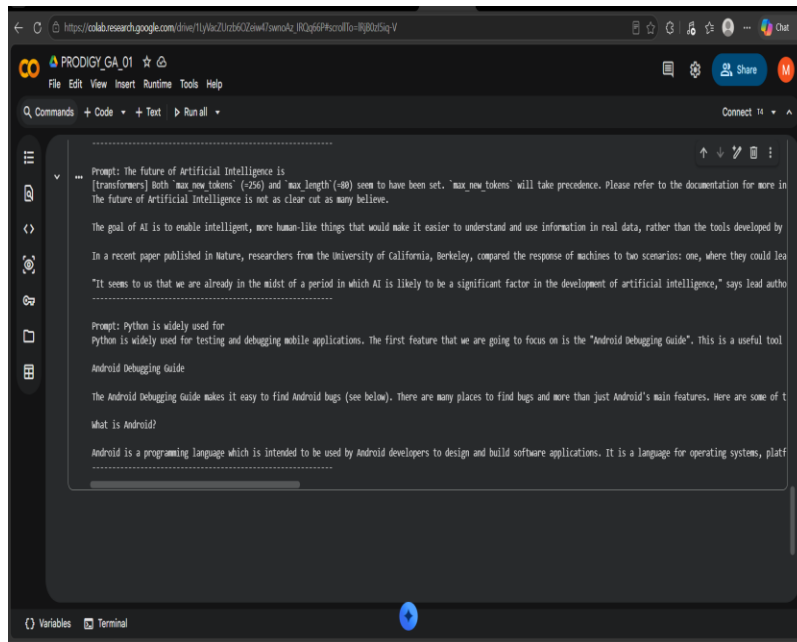
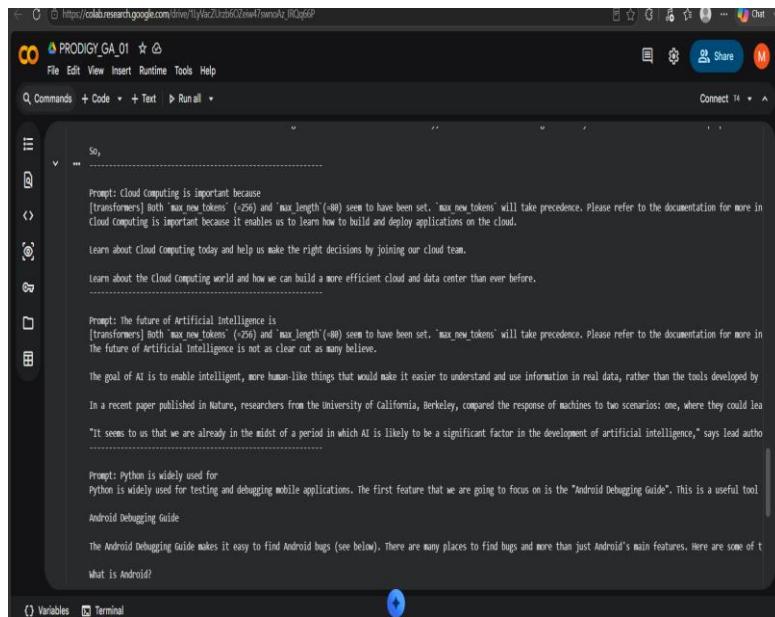
The code cell also shows a warning message from transformers:

```
[transformers] Passing 'generation_config' together with generation-related arguments (['num_return_sequences', 'temperature', 'max_length']) is deprecated and will be removed in a future version. Both 'max_new_tokens' (<256) and 'max_length' (<100) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more info. [transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer GPT2Tokenizer. The clean_up_tokenization post-processing step is designed for WordPiece tokenizer. Artificial Intelligence is transforming the future because it's so far away from reality. That means we'll see it in every job.
```

The generated text is:

```
I asked Dr. Zoraida, and we were both stunned. Dr. Zoraida told me that the next big thing to happen in artificial intelligence will be how we manage these tasks. She said t I asked Dr. Zoraida about the future of AI. She said:
```

We are not going to be able to solve our problems in an artificial way because it will take us a long time. The problem is of the way we want to solve our problems. In order



8. Learning Outcomes:

Learned the fundamentals of GPT-2.

Understood transformer-based language models.

Implemented text generation using Python.

Gained practical experience with Hugging Face Transformers.

Improved knowledge of Generative AI and Natural Language Processing.

9. Conclusion:

This project successfully demonstrated text generation using the GPT-2 pre-trained model. It provided practical experience with transformer-based AI models and enhanced understanding of Generative AI concepts through hands-on implementation.

GitHub Repository Contents

PRODIGY_GA_01.ipynb – Google Colab Notebook

README.md – Project Description

Outputs/ – Execution Screenshots

Final_Documentation.pdf – Project Report

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